

Lesson 22 College Algebra

1) Given: $(x-3)(x+2) = x^2 - x - 6$

$$\frac{x^2 - x - 6}{x - 3} = \underline{\quad ? \quad}$$

2) Given: $(x^2 - 3x + 7)(x - 6) = x^3 - 9x^2 + 25x - 42$

$$\frac{x^3 - 9x^2 + 25x - 42}{x - 6} = \underline{\quad ? \quad}$$

3) Given: $(x^2 - 5x + 1)(x + 7) = x^3 + 2x^2 - 34x + 7$

$$\frac{x^3 + 2x^2 - 34x + 7}{x + 7} = \underline{\quad ? \quad}$$

4) Given: $(x^3 - 8x + 1)(x^2 + x + 7) = x^5 + x^4 - x^3 - 7x^2 - 55x + 7$

$$\frac{x^5 + x^4 - x^3 - 7x^2 - 55x + 7}{x^2 + x + 7} = \underline{\quad ? \quad}$$

5) Given: $(x^4 + x^3 - 8x + 1)(x^2 + x + 7) = x^6 + 2x^5 + 8x^4 - x^3 - 7x^2 - 55x + 7$

$$\frac{x^6 + 2x^5 + 8x^4 - x^3 - 7x^2 - 55x + 7}{x^2 + x + 7} = \underline{\quad ? \quad}$$

6) Use Synthetic Division to divide $\frac{-3x^5 + 5x^3 + 2x + 8}{x - 1}$

a) $\underline{\quad? \quad} \mid -3 \quad \underline{\quad? \quad} \quad \underline{\quad? \quad} \quad \underline{\quad? \quad}$

b) $\frac{-3x^5 + 5x^3 + 2x + 8}{x - 1} = \underline{\quad? \quad}$

7) Use Synthetic Division to divide $\frac{2x^5 + 4x^4 + 2x - 7}{x - 3}$

a) $\underline{\quad? \quad} \mid 2 \quad \underline{\quad? \quad} \quad \underline{\quad? \quad} \quad \underline{\quad? \quad} \quad \underline{\quad? \quad} \quad \underline{\quad? \quad}$

b) $\frac{2x^5 + 4x^4 + 2x - 7}{x - 3} = \underline{\quad? \quad}$

8) Use Synthetic Division to divide $\frac{-2x^4 + 3x^2 - 7}{x + 4}$

a) $\underline{\quad? \quad} \mid -2 \quad \underline{\quad? \quad} \quad \underline{\quad? \quad} \quad \underline{\quad? \quad} \quad \underline{\quad? \quad}$

b) $\frac{-2x^4 + 3x^2 - 7}{x + 4} = \underline{\quad? \quad}$

9) Divide $-4x^4 + 7x^3 - 11x^2 + 2x + 9$ by $x - 2$

a) $\underline{\quad? \quad} \overline{) -4 \quad \quad \quad \underline{\quad? \quad} \quad \quad \underline{\quad? \quad} \quad \quad \underline{\quad? \quad} \quad \quad \underline{\quad? \quad}}$

b) $\frac{-4x^4 + 7x^3 - 11x^2 + 2x + 9}{x - 2} = \underline{\quad? \quad}$

10) Divide $\frac{15x^2 + 12x + 8}{3x - 2}$

Hint:
$$\frac{15x^2 + 12x + 8}{3x - 2} = \frac{(1/3) \cdot 15x^2 + (1/3) \cdot 12x + (1/3) \cdot 8}{(1/3) \cdot 3x - (1/3) \cdot 2}$$
$$= \frac{5x^2 + 4x + 8/3}{x - 2/3}$$

a) $\underline{\quad? \quad} \overline{) 5 \quad \quad \quad \underline{\quad? \quad} \quad \quad \underline{\quad? \quad}}$

b) $\frac{15x^2 + 12x + 8}{3x - 2} = \underline{\quad? \quad}$